

End-to-End Power Automate Flow – Retrieve Microsoft Teams Meeting Transcript using Microsoft Graph API

Architecture Overview

This solution will:

1. Detect Teams meeting invitation emails
2. Extract Teams meeting URL
3. Get Online Meeting ID using Microsoft Graph API
4. Retrieve Transcript ID
5. Download Transcript Content
6. Generate AI Summary
7. Send Email / Store in SharePoint / Teams

Prerequisites

1. Azure App Registration

Go to:

<https://portal.azure.com>

Navigate:

Azure Active Directory → App Registrations → New Registration

App Registration Details

Field	Value
Name	TeamsTranscriptAutomation
Supported Account Type	Single Tenant
Redirect URI	Leave Blank

Click Register.

2. Create Client Secret

Go to:

Certificates & Secrets → New Client Secret

Copy:

- Client ID
- Tenant ID
- Client Secret

Store safely.

3. Add Microsoft Graph API Permissions

Go to:

API Permissions → Add Permission → Microsoft Graph

Choose:

Delegated Permissions

Add:

Permission
OnlineMeetings.Read
OnlineMeetings.ReadWrite
OnlineMeetingTranscript.Read.All
User.Read
Mail.Read

Click:

Grant Admin Consent

Important Note

Transcript APIs only work when:

- Teams meeting recording/transcription is enabled
- User attended or organized meeting
- Transcript generation completed

Power Automate Flow Design

Flow Name

Teams Meeting Transcript Automation

STEP 1 — Trigger

Trigger

When a new email arrives (V3)

Folder

Inbox

Include Attachments

No

Only With Attachments

No

Subject Filter (Optional)

Teams Meeting

STEP 2 — Initialize Variables

Variable 1

Property	Value
Name	JoinWebUrl
Type	String

Variable 2

Property	Value
Name	OnlineMeetingId
Type	String

Variable 3

Property	Value
Name	TranscriptId
Type	String

Variable 4

Property	Value
Name	TranscriptContent
Type	String

STEP 3 — Extract Teams Meeting URL from Email Body

Add Action

HTML to Text

Content

Dynamic Value:

Body

Add Compose Action

Name

Extract Teams Link

Expression

```
first(match(outputs('HTML_to_text'), 'https://teams.microsoft.com/l/meetup-join[^\s]+'))
```

This extracts Teams Join URL from the email body.

STEP 4 — Store Join URL

Set Variable

Variable

JoinWebUrl

Value

Outputs from:

Extract Teams Link

STEP 5 — Get OAuth Token

Add Action

HTTP

Method

POST

URI

```
https://login.microsoftonline.com/TENANT_ID/oauth2/v2.0/token
```

Replace:

TENANT_ID with your Tenant ID.

Headers

```
{  
  "Content-Type": "application/x-www-form-urlencoded"  
}
```

Body

```
client_id=CLIENT_ID&scope=https%3A%2F%2Fgraph.microsoft.com%2F.default&client_secret=CLIENT_SECRET
```

Replace:

- CLIENT_ID
 - CLIENT_SECRET
-

STEP 6 — Parse OAuth Token

Add Parse JSON

Content

```
body('HTTP')
```

Schema

```
{
  "type": "object",
  "properties": {
    "token_type": {
      "type": "string"
    },
    "expires_in": {
      "type": "integer"
    },
    "access_token": {
      "type": "string"
    }
  }
}
```

STEP 7 — Get Online Meeting Details using Join URL

Add HTTP Action

Method

GET

URI

```
https://graph.microsoft.com/v1.0/me/onlineMeetings?$filter=JoinWebUrl eq  
'@{variables('JoinWebUrl')}'
```

Headers

```
{  
  "Authorization": "Bearer ACCESS_TOKEN"  
}
```

Replace ACCESS_TOKEN with:

```
body( 'Parse_JSON' )?[ 'access_token' ]
```

Important

If using Application Permissions:

Use:

```
/users/{userId}/onlineMeetings
```

Instead of:

```
/me/onlineMeetings
```

Example:

```
https://graph.microsoft.com/v1.0/users/user@company.com/onlineMeetings?  
$filter=JoinWebUrl eq 'https://teams.microsoft.com/l/meetup-join/...'
```

STEP 8 — Parse Online Meeting Response

Add Parse JSON

Content

```
body( 'HTTP_2' )
```


Schema

```
{
  "type": "object",
  "properties": {
    "value": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "id": {
            "type": "string"
          }
        }
      }
    }
  }
}
```

STEP 9 — Store Online Meeting ID

Set Variable

Variable

OnlineMeetingId

Value

```
first(body('Parse_JSON_2'))['value']]['id']
```

STEP 10 — Get Transcript List

Add HTTP Action

Method

GET

URI

```
https://graph.microsoft.com/v1.0/me/onlineMeetings/  
@{variables('OnlineMeetingId')}/transcripts
```

Headers

```
{  
  "Authorization": "Bearer ACCESS_TOKEN"  
}
```

STEP 11 — Parse Transcript Response

Add Parse JSON

Content

```
body('HTTP_3')
```

Schema

```
{  
  "type": "object",  
  "properties": {  
    "value": {  
      "type": "array",  
      "items": {  
        "type": "object",  
        "properties": {  
          "id": {  
            "type": "string"  
          }  
        }  
      }  
    }  
  }  
}
```

STEP 12 — Store Transcript ID

Set Variable

Variable

TranscriptId

Value

```
first(body('Parse_JSON_3'))['value']]['id']
```

STEP 13 — Download Transcript Content

Add HTTP Action

Method

GET

URI

```
https://graph.microsoft.com/v1.0/me/onlineMeetings/  
@{variables('OnlineMeetingId')}/transcripts/@{variables('TranscriptId')}/content
```

Headers

```
{  
  "Authorization": "Bearer ACCESS_TOKEN"  
}
```

STEP 14 — Store Transcript Content

Set Variable

Variable

TranscriptContent

Value

```
body('HTTP_4')
```

STEP 15 — Generate AI Summary

Add Action

AI Builder → Create text with GPT

OR

Azure OpenAI

Prompt

```
Summarize the following Teams meeting transcript.
```

```
Provide:
```

1. Meeting Summary
2. Action Items
3. Decisions
4. Risks
5. Follow Ups

```
Transcript:
```

```
@{variables('TranscriptContent')}
```

STEP 16 — Send Summary Email

Add Action

Send an Email (V2)

To

Manager or Distribution List

Subject

Teams Meeting Summary

Body

```
<h2>Meeting Summary</h2>  
  
@{outputs('Create_text_with_GPT')}
```

OPTIONAL — Store Transcript in SharePoint

Add Action

Create File

Site Address

Your SharePoint Site

Folder Path

/Shared Documents/Transcripts

File Name

```
MeetingTranscript.txt
```

File Content

```
variables('TranscriptContent')
```

OPTIONAL — Post Summary to Teams

Add Action

Post message in a chat or channel

Message

```
Meeting Summary:  
  
@{outputs('Create_text_with_GPT')}
```

FULL GRAPH API FLOW

1. Get OAuth Token

```
POST  
https://login.microsoftonline.com/{tenantId}/oauth2/v2.0/token
```

2. Get Online Meeting

```
GET  
https://graph.microsoft.com/v1.0/me/onlineMeetings?$filter=JoinWebUrl eq  
'{JoinUrl}'
```

3. Get Transcript List

GET

`https://graph.microsoft.com/v1.0/me/onlineMeetings/{meetingId}/transcripts`

4. Download Transcript

GET

`https://graph.microsoft.com/v1.0/me/onlineMeetings/{meetingId}/transcripts/{transcriptId}/content`

Common Errors and Fixes

Error

401 Unauthorized

Fix

- Verify Access Token
 - Verify Graph Permissions
 - Grant Admin Consent
-

Error

403 Forbidden

Fix

- Missing transcript permissions
 - Transcript not enabled
-

Error

Meeting Not Found

Fix

- Join URL encoding issue
 - User mismatch
 - Meeting belongs to another organizer
-

Best Practice

Use Application Permissions + Application Access Policy for enterprise solutions.

This allows:

- Access to all users' meetings
 - Background automation
 - No manual login required
-

Recommended Enterprise Architecture

Flow 1

Email Trigger → Store Meeting Metadata

Flow 2

Scheduler Flow → Retrieve Transcript

Flow 3

AI Summary Flow

Flow 4

Archive to SharePoint + Dataverse

Final Output Example

Meeting Summary

- Discussed Power Platform Governance

- Reviewed SPFx deployment
- Approved production migration

Action Items

- Ravi to deploy SPFx package
- Infra team to approve API permissions
- Security team to validate Azure App Registration

Risks

- Missing Graph permissions
 - Transcript delay after meeting
-

Advanced Enhancements

You can further enhance solution using:

- Dataverse storage
 - Power BI dashboards
 - Azure Functions
 - Adaptive Cards
 - Copilot Studio
 - SharePoint Knowledge Base
 - Semantic Search
 - RAG architecture
-

Production Recommendation

For production:

- Use child flows
 - Store secrets in Azure Key Vault
 - Use environment variables
 - Add retry policies
 - Add exception handling
 - Add transcript polling logic
 - Enable logging
-

Transcript Polling Logic

Sometimes transcript generation takes 5–15 minutes.

Use:

Do Until Loop

Condition:

```
length(body('HTTP_3')['value']) > 0
```

Delay:

2 Minutes

Timeout:

PT30M

Final Result

This automation fully:

✓ Retrieves Teams Meeting URL ✓ Finds Online Meeting ID ✓ Retrieves Transcript ID ✓ Downloads Transcript Content ✓ Generates AI Summary ✓ Sends Notifications ✓ Stores Records ✓ Works End-to-End using Graph API + Power Automate